

Homework 1

Due : Friday, September 19 at 11 : 59 pm

Deliverables. Submit a single PDF of your write-up to Gradescope *HW1 Write-Up*.

Start each problem on a new page.

Honor Code

Write and sign the following statement:

“I certify that all solutions in this document are entirely my own and that I have not looked at anyone else’s solution. I have given credit to all external sources I consulted.”

Since time is valuable and this is a non-credit independent study, most of the typesetting and English translation here were produced by Cursor based on my handwritten scripts. I’m grateful for the help that AI has provided to my learning.

Written By Ziyu Li on Dec15, 2025

Linear Algebra

1. **System of equations (3 points).** Consider the linear system

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 12 \end{bmatrix}.$$

How many solutions does this system have? Explain your reasoning.

$$\text{Let } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$$\text{Let } B = \begin{bmatrix} 1 & 2 & 3 & 6 \\ 2 & 0 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

$\text{Rank}(A) = 2$ and $\text{Rank}(B) = 3$, so there is no solution.

2. **Asymptotic powers of 2×2 matrices (6 points).** For each matrix below, determine the behavior of

$$\lim_{n \rightarrow \infty} M^n$$

Hint: Use the eigen-decomposition $M = PDP^{-1}$, where $D = \{\lambda_1, \lambda_2, \dots\}$ is a diagonal matrix of the eigenvalues. What would be the formula for M^n ?

(a) $M_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$

(b) $M_2 = \begin{bmatrix} 2 & -5 \\ 1/2 & -7/6 \end{bmatrix}.$

(c) $M_3 = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}.$

answer to (a): M_1 has two eigenvalues 1 and -1. let:

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

So:

$$M = P \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} P^{-1}$$

$$\lim_{n \rightarrow \infty} M^n = P \begin{pmatrix} 1 & 0 \\ 0 & (-1)^n \end{pmatrix} P^{-1}$$

answer to (b): M_1 has two eigenvalues $\frac{1}{2}$ and $\frac{1}{3}$. let:

$$P = \begin{pmatrix} 10 & 3 \\ 3 & 1 \end{pmatrix}$$

So:

$$M = P \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} P^{-1}$$

$$\lim_{n \rightarrow \infty} M^n = P \begin{pmatrix} (\frac{1}{2})^n & 0 \\ 0 & (\frac{1}{3})^n \end{pmatrix} P^{-1} = 0$$

answer to (c): M_1 has two eigenvalues 2 and 3. let:

$$P = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$$

So:

$$M = P \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} P^{-1}$$

$$\lim_{n \rightarrow \infty} M^n = P \begin{pmatrix} 2^n & 0 \\ 0 & 3^n \end{pmatrix} P^{-1}$$

3. **Singular Value Decomposition (4 points).** Given the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$$

Find a unit vector x ($\|x\| = 1$) for which $\|Ax\|$ is maximized.

Hint: Recall from SVD that $A = U\Sigma V^\top$, where A maps each right singular vector (a column of V) to a left singular vector (a column of U), scaled by the corresponding singular value. For which right singular vector does this scaling make $\|Ax\|$ the largest?

Consider $\|Ax\|^2 = x^T A^T A x$, since $\|x\| = 1$ so let $\lambda_m = \max(\lambda_{A^T A})$ and $x = [0 \dots 1 \dots 0]$ where 1 lies on the m index. Now it reaches its maximum value λ_m

The eigenvalues of $A^T A$ are 9 and 4, so $\|Ax\|$ reach maximum value $\sqrt{9} = 3$ when $x = [\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}]$, which is the eigen vector of $A^T A$

- (a) What is the size of the transformation matrix T that performs this flip? *Hint: You can first convert I to a 1×9 vector, make transformation using matrix T , and then convert I_{flip} back to a 3×3 matrix.*

Answer. We first flatten the 3×3 matrix I into a 1×9 row vector and multiply it on the right by T . In order for $I_{\text{row}}T$ to still be a 1×9 row vector, the matrix T must be of size

$$T \in \mathbb{R}^{9 \times 9}.$$

- (b) Construct T and verify that $I \times T$ produces I_{flip} .

Answer. Flatten I in row-major order as

$$I_{\text{row}} = [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9].$$

The flipped image (along the Y axis, i.e. top and bottom rows are swapped) corresponds to

$$I_{\text{flip, row}} = [x_7, x_8, x_9, x_4, x_5, x_6, x_1, x_2, x_3].$$

Thus we want a permutation matrix $T \in \mathbb{R}^{9 \times 9}$ such that

$$I_{\text{row}}T = I_{\text{flip, row}}.$$

One such matrix T is

$$T = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

One can check directly (column by column) that

$$[x_1, \dots, x_9]T = [x_7, x_8, x_9, x_4, x_5, x_6, x_1, x_2, x_3],$$

and reshaping this row vector back into a 3×3 matrix gives exactly I_{flip} .

- (c) Describe an algorithm for constructing the vertical-flip transformation matrix for any $N \times N$ matrix (either text explanation or pseudocode is acceptable). Show that $I \times T$ produces I_{flip} .

Answer. Consider an $N \times N$ image I . Flatten it in row-major order into a $1 \times N^2$ row vector

$$v = [v_0, v_1, \dots, v_{N^2-1}],$$

where index k (zero-based) corresponds to the pixel at row

$$r = \left\lfloor \frac{k}{N} \right\rfloor, \quad c = k \bmod N.$$

A vertical flip (along the Y axis) reverses the row index but keeps the column index:

$$(r, c) \mapsto (r', c') = (N - 1 - r, c).$$

The new (flattened) index after the flip is

$$j = r' \cdot N + c' = (N - 1 - r)N + c.$$

We construct $T \in \mathbb{R}^{N^2 \times N^2}$ as a permutation matrix by setting

$$T_{k+1, j+1} = 1$$

for each $k = 0, \dots, N^2 - 1$ (and all other entries of T are zero). Then for the row vector v we have

$$(vT)_j = v_k,$$

meaning that the pixel originally at row r and column c moves to row $N - 1 - r$ and column c in the reshaped $N \times N$ matrix. Therefore vT corresponds exactly to the vertically flipped image I_{flip} .

- (d) Describe how you would modify this algorithm to do a *horizontal* flip (along the X axis).

Answer. For a horizontal flip (along the X axis), we keep the row index and flip the column index. Using the same notation as above, the mapping becomes

$$(r, c) \mapsto (r', c') = (r, N - 1 - c).$$

The new flattened index is

$$j = r \cdot N + (N - 1 - c).$$

We again construct a permutation matrix $T \in \mathbb{R}^{N^2 \times N^2}$ by setting

$$T_{k+1, j+1} = 1$$

for each $k = 0, \dots, N^2 - 1$ (all other entries are zero). Then vT corresponds to reversing the columns in each row, which is exactly a horizontal flip of the original $N \times N$ image.

Calculus

5. Partial Derivatives (8 points). (First and second derivatives)

- (a) For $f(x_1, x_2) = x_1^3 + x_2^3 - 3x_1x_2$, find all the first and second order partial derivatives.

Answer. First-order partial derivatives:

$$\frac{\partial f}{\partial x_1} = 3x_1^2 - 3x_2, \quad \frac{\partial f}{\partial x_2} = 3x_2^2 - 3x_1.$$

Second-order partial derivatives:

$$\frac{\partial^2 f}{\partial x_1^2} = 6x_1, \quad \frac{\partial^2 f}{\partial x_2^2} = 6x_2, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = -3.$$

- (b) Let $f(x, y) = 4x^2 + y^2 - 8xy + 4x + 6y - 10$.

Find the critical points by solving

$$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

simultaneously and determine which point(s) yield a minimum value.

Answer. First-order partial derivatives:

$$\frac{\partial f}{\partial x} = 8x - 8y + 4, \quad \frac{\partial f}{\partial y} = 2y - 8x + 6.$$

Solve

$$8x - 8y + 4 = 0, \quad 2y - 8x + 6 = 0.$$

From the first equation,

$$8x - 8y + 4 = 0 \Rightarrow x - y + \frac{1}{2} = 0 \Rightarrow x = y - \frac{1}{2}.$$

Plug into the second:

$$2y - 8(y - \frac{1}{2}) + 6 = 0 \Rightarrow 2y - 8y + 4 + 6 = 0 \Rightarrow -6y + 10 = 0 \Rightarrow y = \frac{5}{3},$$

hence

$$x = y - \frac{1}{2} = \frac{5}{3} - \frac{1}{2} = \frac{7}{6}.$$

So the only critical point is $(\frac{7}{6}, \frac{5}{3})$.

To classify it, compute the Hessian:

$$f_{xx} = 8, \quad f_{yy} = 2, \quad f_{xy} = f_{yx} = -8.$$

The determinant of the Hessian is

$$D = f_{xx}f_{yy} - f_{xy}^2 = 8 \cdot 2 - (-8)^2 = 16 - 64 = -48 < 0.$$

Since $D < 0$, the Hessian is indefinite, so the critical point $(\frac{7}{6}, \frac{5}{3})$ is a saddle point, not a minimum. Therefore, *there is no point that yields a (local) minimum value.*

- (c) For $f(x, y) = e^{xy} + x^2y$, compute:

- (i)
- $\frac{\partial f}{\partial x}$
- and
- $\frac{\partial f}{\partial y}$

Answer.

$$\frac{\partial f}{\partial x} = ye^{xy} + 2xy, \quad \frac{\partial f}{\partial y} = xe^{xy} + x^2.$$

- (ii)
- $\frac{\partial^2 f}{\partial x^2}$
- ,
- $\frac{\partial^2 f}{\partial y^2}$
- , and
- $\frac{\partial^2 f}{\partial x \partial y}$

Answer.

$$\frac{\partial^2 f}{\partial x^2} = y^2 e^{xy} + 2y,$$

$$\frac{\partial^2 f}{\partial y^2} = x^2 e^{xy},$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = e^{xy} + xye^{xy} + 2x.$$

- (d) For
- $f(x, y) = \ln(x^2 + y^2 + 1)$
- , compute:

- (i)
- $\frac{\partial f}{\partial x}$
- and
- $\frac{\partial f}{\partial y}$

Answer. Let $g(x, y) = x^2 + y^2 + 1$. Then

$$\frac{\partial f}{\partial x} = \frac{1}{g(x, y)} \cdot 2x = \frac{2x}{x^2 + y^2 + 1},$$

$$\frac{\partial f}{\partial y} = \frac{1}{g(x, y)} \cdot 2y = \frac{2y}{x^2 + y^2 + 1}.$$

- (ii)
- $\frac{\partial^2 f}{\partial x^2}$
- ,
- $\frac{\partial^2 f}{\partial y^2}$
- , and
- $\frac{\partial^2 f}{\partial x \partial y}$

Answer. Using $g(x, y) = x^2 + y^2 + 1$,

$$\frac{\partial^2 f}{\partial x^2} = \frac{2}{g(x, y)} - \frac{4x^2}{g(x, y)^2} = \frac{2(x^2 + y^2 + 1) - 4x^2}{(x^2 + y^2 + 1)^2} = \frac{2(-x^2 + y^2 + 1)}{(x^2 + y^2 + 1)^2},$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{2}{g(x, y)} - \frac{4y^2}{g(x, y)^2} = \frac{2(x^2 + y^2 + 1) - 4y^2}{(x^2 + y^2 + 1)^2} = \frac{2(x^2 - y^2 + 1)}{(x^2 + y^2 + 1)^2},$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y} \left(\frac{2x}{x^2 + y^2 + 1} \right) = -\frac{4xy}{(x^2 + y^2 + 1)^2}.$$

6. Recursive expression and derivatives (6 points).

Suppose we have variables $z_1, z_2, \dots, z_n$, $w_1, \dots, w_{n-1}$, and $b_1, \dots, b_{n-1} \in \mathbb{R}$, where

$$z_n = w_{n-1}z_{n-1} + b_{n-1}.$$

- (a) Compute $\frac{dz_k}{dz_{k-1}}$ for $k = 2, \dots, n$.

Answer. From the recursive relation

$$z_k = w_{k-1}z_{k-1} + b_{k-1},$$

treating w_{k-1}, b_{k-1} as constants when differentiating with respect to z_{k-1} , we obtain

$$\frac{dz_k}{dz_{k-1}} = w_{k-1}, \quad \text{for } k = 2, \dots, n.$$

- (b) Compute $\frac{dz_n}{dz_1}$.

Answer. By the chain rule,

$$\frac{dz_n}{dz_1} = \frac{dz_n}{dz_{n-1}} \cdot \frac{dz_{n-1}}{dz_{n-2}} \cdots \frac{dz_2}{dz_1}.$$

Using part (a), we have $\frac{dz_k}{dz_{k-1}} = w_{k-1}$, so

$$\frac{dz_n}{dz_1} = \prod_{j=1}^{n-1} w_j.$$

- (c) Compute $\frac{dz_n}{db_1}$.

You may express your answers in terms of $z_1, \dots, z_n, w_1, \dots, w_{n-1}, b_1, \dots, b_{n-1}$.

Answer. The bias b_1 first appears in

$$z_2 = w_1z_1 + b_1,$$

so

$$\frac{dz_2}{db_1} = 1.$$

For $k \geq 3$, since $z_k = w_{k-1}z_{k-1} + b_{k-1}$, by the chain rule

$$\frac{dz_k}{db_1} = \frac{dz_k}{dz_{k-1}} \cdot \frac{dz_{k-1}}{db_1} = w_{k-1} \frac{dz_{k-1}}{db_1}.$$

Iterating this from $k = 3$ to $k = n$ gives

$$\frac{dz_n}{db_1} = w_{n-1}w_{n-2} \cdots w_2 \cdot \frac{dz_2}{db_1} = \prod_{j=2}^{n-1} w_j.$$

(For the special case $n = 2$, the product from $j = 2$ to 1 is an empty product, which we interpret as 1, consistent with $z_2 = w_1z_1 + b_1$.)

Probability

7. **Conditioned uniform difference (3 points).** Let $X, Y \stackrel{\text{iid}}{\sim} \text{Unif}(-1, 1)$. Compute

$$\mathbb{P}(|X - Y| \leq 0.5 \mid XY > 0).$$

Hint: Try drawing a 2D cartesian plane where the horizontal and vertical axes represent X and Y respectively and each range from -1 to 1. For which quadrants is it true that $XY > 0$? Within these quadrants, how can we visualize the region $|X - Y| < 0.5$?

Answer. Since $X, Y \stackrel{\text{iid}}{\sim} \text{Unif}(-1, 1)$, the pair (X, Y) is uniform on the square $[-1, 1] \times [-1, 1]$ with total area 4.

The event $\{XY > 0\}$ means X and Y have the same sign, so this corresponds to the union of the first and third quadrants inside the square:

$$\{(x, y) : 0 < x < 1, 0 < y < 1\} \cup \{(x, y) : -1 < x < 0, -1 < y < 0\},$$

each of area 1, so

$$\text{Area}(XY > 0) = 2, \quad \mathbb{P}(XY > 0) = \frac{2}{4} = \frac{1}{2}.$$

Now consider the region $\{|X - Y| \leq 0.5\}$ within these quadrants. In the first quadrant, we look at the unit square $[0, 1] \times [0, 1]$ and the band

$$|x - y| \leq 0.5.$$

This band is the unit square minus two isosceles right triangles where $|x - y| > 0.5$: each triangle has side length $1 - 0.5 = 0.5$ and area $\frac{1}{2}(0.5)^2 = 0.125$, so the total area of the two triangles is 0.25. Hence, in the first quadrant

$$\text{Area}(|x - y| \leq 0.5) = 1 - 0.25 = 0.75.$$

By symmetry, in the third quadrant $(-1, 0) \times (-1, 0)$ the area of the region $\{|x - y| \leq 0.5\}$ is also 0.75. Therefore

$$\text{Area}(|X - Y| \leq 0.5, XY > 0) = 0.75 + 0.75 = 1.5.$$

Thus

$$\mathbb{P}(|X - Y| \leq 0.5, XY > 0) = \frac{1.5}{4} = \frac{3}{8}.$$

Finally, by the definition of conditional probability,

$$\mathbb{P}(|X - Y| \leq 0.5 \mid XY > 0) = \frac{\mathbb{P}(|X - Y| \leq 0.5, XY > 0)}{\mathbb{P}(XY > 0)} = \frac{\frac{3}{8}}{\frac{1}{2}} = \frac{3}{4}.$$

So

$$\mathbb{P}(|X - Y| \leq 0.5 \mid XY > 0) = \frac{3}{4}.$$

8. **Nearest-neighbor arc length (4 points).** Suppose you select 20 i.i.d. points $X_1, \dots, X_{20}$ uniformly at random on the circumference of the unit circle.

- i. Let D be the shortest arc distance from X_1 to the nearest of the other 19 points. Calculate $\mathbb{P}(D > t)$ where $0 \leq t \leq \frac{1}{2}$.

Answer. Since $X_1, \dots, X_{20}$ are i.i.d. Unif on the circumference of the unit circle, we can parametrize the circle by arc length in $[0, 1)$ (total circumference normalized to 1). By rotational symmetry, we may fix X_1 at some point and measure all distances relative to it.

The event $\{D > t\}$ means that *every* one of the other 19 points is at arc distance strictly greater than t from X_1 . On the circle, the set of points whose distance from X_1 is at most t is an arc of total length $2t$ (an interval of length t on each side of X_1). For any single point X_j ($j \geq 2$),

$$\mathbb{P}(X_j \text{ lies outside this } 2t\text{-arc}) = 1 - 2t.$$

Since the 19 points are independent,

$$\mathbb{P}(D > t) = (1 - 2t)^{19}, \quad 0 \leq t \leq \frac{1}{2}.$$

- ii. Find the expected arc length (in degrees) between X_1 and the point nearest to it.

Answer. The random variable D takes values in $[0, 1/2]$, and we have

$$\mathbb{P}(D > t) = (1 - 2t)^{19}, \quad 0 \leq t \leq \frac{1}{2}.$$

Using the tail-integral formula for a nonnegative continuous random variable,

$$\mathbb{E}[D] = \int_0^\infty \mathbb{P}(D > t) dt = \int_0^{1/2} (1 - 2t)^{19} dt.$$

Let $u = 1 - 2t$. Then $du = -2dt$ so $dt = -\frac{1}{2}du$, and the limits change from $t = 0 \Rightarrow u = 1$ to $t = \frac{1}{2} \Rightarrow u = 0$. Thus,

$$\mathbb{E}[D] = \int_0^{1/2} (1 - 2t)^{19} dt = \int_1^0 u^{19} \left(-\frac{1}{2} du\right) = \frac{1}{2} \int_0^1 u^{19} du = \frac{1}{2} \cdot \frac{1}{20} = \frac{1}{40}.$$

Here D is measured as a fraction of the full circle (total circumference 1). Since a full circle corresponds to 360° , the expected arc length in degrees is

$$\mathbb{E}[\text{arc length in degrees}] = \frac{1}{40} \times 360^\circ = 9^\circ.$$

9. **Cancer screening (3 points).** A medical test has sensitivity 90% and false-positive rate 3%:

$$\mathbb{P}(T = 1 \mid C = 1) = 0.9, \quad \mathbb{P}(T = 1 \mid C = 0) = 0.03.$$

Suppose the disease prevalence is very low, $\mathbb{P}(C = 1) = 0.001$. Compute the posterior probability of disease given a positive test:

$$\mathbb{P}(C = 1 \mid T = 1).$$

Answer. We apply Bayes' rule:

$$\mathbb{P}(C = 1 \mid T = 1) = \frac{\mathbb{P}(T = 1 \mid C = 1)\mathbb{P}(C = 1)}{\mathbb{P}(T = 1)}.$$

The marginal probability of a positive test is

$$\mathbb{P}(T = 1) = \mathbb{P}(T = 1 \mid C = 1)\mathbb{P}(C = 1) + \mathbb{P}(T = 1 \mid C = 0)\mathbb{P}(C = 0).$$

Using $\mathbb{P}(T = 1 \mid C = 1) = 0.9$, $\mathbb{P}(T = 1 \mid C = 0) = 0.03$ and $\mathbb{P}(C = 1) = 0.001$ (so $\mathbb{P}(C = 0) = 0.999$), we get

$$\mathbb{P}(T = 1) = 0.9 \cdot 0.001 + 0.03 \cdot 0.999 = 0.0009 + 0.02997 = 0.03087.$$

Therefore,

$$\mathbb{P}(C = 1 \mid T = 1) = \frac{0.9 \cdot 0.001}{0.03087} = \frac{0.0009}{0.03087} \approx 0.029 \approx 2.9\%.$$

So even with a positive test result, the posterior probability of having the disease is only about 2.9%.

10. **Follower Counts (3 points).** Suppose 420 people are sitting uniformly at random around a circle, each with a *distinct* number of TikTok followers. What is the expected number of people whose follower count is higher than both their immediate neighbors?

Answer. Label the people on the circle as $1, 2, \dots, 420$, and for each person i consider the three people: person i and their two immediate neighbors. Since all follower counts are distinct and the seating is uniform at random, the relative order of the three follower counts is equally likely among the $3! = 6$ permutations. Therefore, each of the three has probability $1/3$ to be the one with the highest follower count among the three.

Define the indicator

$$I_i = \begin{cases} 1, & \text{if person } i \text{ has more followers than both neighbors,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{P}(I_i = 1) = \frac{1}{3}, \quad \text{for each } i = 1, \dots, 420.$$

Let $S = \sum_{i=1}^{420} I_i$ be the total number of people whose follower count is higher than both neighbors. By linearity of expectation,

$$\mathbb{E}[S] = \sum_{i=1}^{420} \mathbb{E}[I_i] = \sum_{i=1}^{420} \mathbb{P}(I_i = 1) = 420 \cdot \frac{1}{3} = 140.$$

Hence, the expected number of such people is 140.